
The Re-Emergence of Emotional Appeals in Interactive Data Visualization

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The Re-Emergence of Emotional Appeals in Interactive Data Visualization

By Charles Kostelnick

Abstract

Purpose: I argue that emotional appeals, prevalent in charts and graphs during the later nineteenth century but largely dormant since then, have rapidly re-emerged in contemporary data visualization. Changing the relationship between designer and user, this new form of data design has intensified the affective impact of data displays by eliciting emotions ranging from excitement and empathy to anxiety and fear.

Methods: This article draws on historical and contemporary sources to build its case. It gives an overview of emotional appeals in the rhetorical tradition, from Aristotle to modern theorists like George Campbell, who emphasized sensory responses through personalization and proximity. The article provides an historical overview of pathos appeals in data design during the later nineteenth century and the shift to modernist minimalism in the twentieth century. Contemporary examples from companies, nonprofits, government agencies, and individual designers illustrate how data visualization arouses emotion.

Results: Emotional appeals during the nineteenth century focused primarily on color and design novelty, which, by appealing largely to the senses, fostered emotional responses such as excitement and curiosity. Contemporary data visualization makes similar emotional appeals through the use of color, novelty, and multimodal features; however, digital technology also allows designers to appeal to the emotions by personalizing displays through interactivity, spatial and temporal proximity, and aesthetic and expressive elements.

Conclusion: Pathos (emotional) appeals have become an integral part of contemporary data visualization, largely because of the multimodal and interactive affordances of digital technology. Designers who understand this dimension of data design can deploy technology to make their displays more engaging, humane, and usable.

Keywords: Visual communication, data visualization, charts and graphs, visual rhetoric, history of data design

Practitioner's Takeaway:

- Although emotional elements in data design are often avoided in favor of concise objectivity, they have a long history and are becoming increasingly prevalent online.
- Deploying color and multimodal features creates visual stimuli that arouse users emotionally, creating excitement and enhancing user engagement.
- Adding interactive features enables users to customize the data and the data's graphical design, fostering both usability and a stronger emotional attachment to the display.
- Visualizing real-time (or fluid) data increases the user's temporal proximity to the display, heightening its emotional impact.
- Affording users opportunities for feedback and discussion invests them emotionally in the design by building community and enhancing interest and understanding.

We stand in the midst of an explosion in data visualization that rivals the “golden age” of the nineteenth century (Funkhouser, 1937, p. 330; Friendly, 2008) in invention and ingenuity and that far exceeds that era in its capacity to make data accessible to massive audiences. With trillions of charts and graphs now generated annually (Tufte, 1990, p. 9), data design has become a major activity in business, technical, and professional communication. Interactive data visualizations now appear all across the Internet, generated by government agencies, news outlets, private companies, nonprofits, and a growing multitude of freelance digital designers (see McCandless, 2010). These digital charts and graphs offer new design dimensions that differ radically from static displays. The data sets are frequently large and complex, containing thousands of data points, and are often dynamic rather than static; they typically allow users to shape or alter displays, redefining agency (Rawlins & Wilson, 2014), and even allow users to interact with designers and other audience members. And the growing array of visualization forms often contain novel, multimodal, or hybrid elements, which can engage and delight users as well as challenge and stretch their interpretive skills.

Surprisingly, given these profound and rapid changes in data design, scant attention has been given to how these new and potent forms are reshaping the user's experience emotionally—of how they are humanizing data design to engage and persuade audiences. This reintroduction of emotional elements is both startling and predictable: after playing a large role during the golden age of statistical graphics, pathos appeals had largely been avoided in serious data design over the past century, as modernist minimalism and other factors conspired to leach them out, relegating them mostly to mass media infographics laden with what Tufte famously demonized as “chartjunk” (1983, pp. 107–121). With the advent of online interactive graphics, however, emotional appeals have now been re-established and tacitly legitimized (see Kostelnick, 2008).

Here, I will argue that pathos appeals are intrinsic to the new digital media by way of design elements and through user affordances that make data visualization even possible. This infusion of emotion happens in two ways—one a revival and the other largely new:

1. Data visualization revives the nineteenth-century profusion of *sensory stimuli* through design features

like color, multimodality, animation, pictorial elements, and novelty—all of which arouse emotional responses.

2. Data visualization personalizes data design by developing a close proximity between data and audience. It accomplishes this personalizing effect by:
 - enabling users to shape and scale data displays to meet their needs;
 - affording interaction with other users and designers;
 - visualizing fluid (often real-time) data, heightening the immediacy of the data; and
 - encouraging self-expression, including the visualization of personal data.

Data designers can deploy these and other digital features to arouse users emotionally, creating excitement and enhancing user engagement and understanding. Although emotion in data design has been debunked in favor of concise objectivity and cognitive efficiency, emotion should less be loathed and evaded than acknowledged, deployed, and controlled. By exploiting this opportunity to connect with their audiences emotionally, designers can humanize their displays, drawing on a long, foundational tradition from which they can glean both precedent and inspiration.

Emotional (Pathos) Appeals in Historical Context

Although emotional appeals have historically received far less attention than the study of rational appeals (Waddell, 1990, pp. 381–382, 397), the concept of pathos has had an enduring role in the study of rhetoric. Arousing emotional responses in verbal language begins at least with Aristotle, who views emotion from two different perspectives: on the one hand, as a possible barrier to rational thinking that undermines logic with overly subjective responses and on the other, as a key element of persuasion that powerfully engages audiences (Jasinski, 2001, p. 421). Regarding the latter perspective, in which emotions enhance arguments, Aristotle (2007) defines and analyzes emotions rhetorically as “those things through which, by undergoing change, people come to differ in their judgments and which are accompanied by pain and pleasure, for example, anger, pity, fear, and other such things and their opposites” (p. 113). For example, in the pain/pleasure dichotomy, anger is the

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opposite of calmness, enmity that of friendliness, fear that of confidence, and so on (Aristotle, pp. 116–147). The speaker seeks to evoke and manage these emotions in trying to persuade an audience in a judicial proceeding or other forum, an art that Roman orators like Cicero and Quintilian greatly expanded and refined (Katula, 2003).

The applications and attributes of emotional appeals have been addressed by many modern rhetoricians, probably most extensively by the Enlightenment philosopher and theologian George Campbell. In his *Philosophy of Rhetoric* (1776), Campbell establishes a central role for emotion in constructing persuasive speeches and sermons: “To say, that it is possible to persuade without speaking to the passions, is but at best a kind of specious nonsense” (p. 199). Passion “animates” our ideas and gives them “spirit and energy” (p. 199). In short, the emotional (“pathetic”) and the rational must go hand in hand to construct persuasive speeches (p. 201). Echoing Edmund Burke’s 1757 treatise on the *Sublime and Beautiful*, Campbell claims that the senses have the strongest ability to arouse passion. “A passion is most strongly excited by sensation” (pp. 207–208), claims Campbell, followed by memory and imagination, the last of which fosters the first two (pp. 207–208). Campbell describes the “circumstances” by which imagination stimulates the emotions: “probability, plausibility, importance, proximity of time, connexion of place, relation of the actors or sufferers to the hearers or speaker, interest of the hearers or speaker in the consequences” (p. 209). Some aspects of emotion, then, pertain to “proximity of time,” especially the immediate past or immediate future (pp. 221–224), and the “connexion of place” (pp. 224–226). In other words, we usually experience stronger feelings relative to those people or situations closest to us.

For example, Campbell observes that when we read about “accidents” we are more concerned (and moved emotionally) by one that occurs “in our neighborhood” than “the most deplorable accidents in countries distant and unknown” (p. 226). The most potent circumstance, Campbell argues, is “interest in the consequences” (pp. 227–229) because here the audience sees itself as having a personal stake in the subject, even if only vicariously. This personal aspect of pathos—often fostered by the close proximity of space and time—plays a central role in the emotional elements that I will explore in contemporary data visualization.

The physiological and psychological aspects of emotion, often prompted by sensory experience, still

infuse contemporary rhetoric. However, the traditional separation of emotion and reason has somewhat diminished with contemporary rhetorical scholarship (Jasinski, 2001, pp. 424–425). Some contemporary theorists view emotion through a cognitive framework that encompasses the ability to reason and that, therefore, dovetails with a rational (logos) response (Jasinski, 2001, pp. 424–425; see also Katula, 2003, pp. 5–6). In short, emotion can complement rather than contradict logical analysis, an important theoretical insight for data design where emotion and reason are often placed at odds.

Emotional Elements in the Golden Age of Statistical Graphics

Like other design forms—painting, architecture, sculpture, graphic design—data design also has a history of invoking pathos appeals, especially during the “golden age” (Funkhouser, 1937; Friendly, 2008) when Romantic and Victorian values dominated all forms of design. During this period, in the second half of the of the nineteenth century, a rich variety of new graphical forms appeared, building on innovations of William Playfair (1801) and other pioneers (see Friendly & Denis, 2001, for a detailed timeline of significant examples). Charles Joseph Minard’s famous flow chart of Napoleon’s Russian campaign (1878; see also Tufte, 1983, p. 41); Florence Nightingales’s area charts for documenting conditions in military hospitals (see Brasseur 2005); Charles Booth’s poverty maps of London (1902); and the panoply of charts, graphs, and maps in the U.S. statistical atlases (Walker, 1874; Hewes & Gannett, 1883; Gannett, 1898) and in the French national statistical albums (Ministère des Travaux Publics)—all contributed to the growing array of forms available to professionals and the public. This new medium for representing quantitative information engendered emotion through sensory experience, novelty, and excitement, with a combination of physical and psychological stimuli that engaged the reader’s eye, brain, and heart. Graphical elements that evoked pathos appeals included lavish color, pictorial elements, innovative new genres, and epic narratives.

Abundant use of color

When it burst forth in late nineteenth-century data design, color stirred the emotions, both perceptually and

culturally. The printing technology of chromolithography supported the inexpensive reproduction of color, and designers took full advantage of it. The U.S. statistical atlases used a full palette of colors to visualize census data, shown in Figure 1 in the area chart and accompanying map from the 1898 *Statistical Atlas of the United States* (Gannett, Plate 22). This huge chart shows the relative size of population groups from the founding of the country to the present, with the "Native Stock" in the center a light, flesh-colored tone and various other segments (European foreigners on the left, African Americans on the right) in darker shades, notably a darker shade of pink for British (suggesting the greatest affinity with the native stock), dark green for the Irish, and light brown for African Americans. This lavish use of

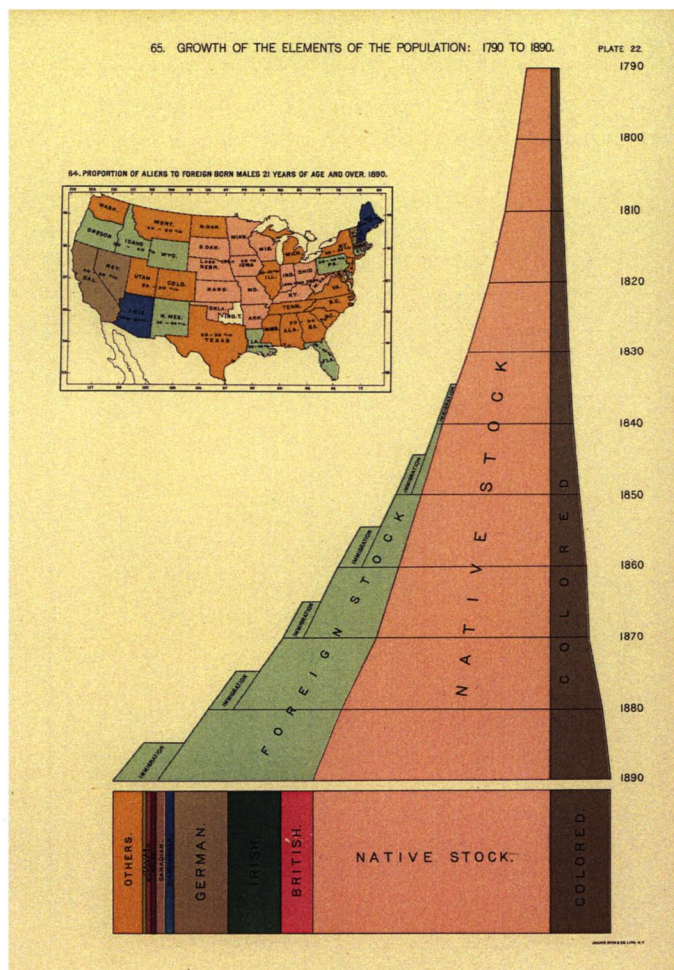


Figure 1. Area chart and map from the 1898 *Statistical Atlas of the United States* showing segments of the population for each census since 1790 (Gannett, Plate 22). Courtesy of the Library of Congress, Geography and Map Division.

color reflects not only the cultural and political attitudes of the era but also Victorian values about the capacity of design to arouse emotion through sensory stimulation (see Kimball, 2006).

Novel/playful breaking of boundaries

Many nineteenth-century designs introduced novel design elements, creating excitement and prompting discovery. Design conventions were still in a state of flux, as designers experimented with new forms, like the massive area chart in Figure 1, which has few predecessors and few progeny. Even quotidian data can arouse curiosity and foster engagement when visualized in novel, colorful ways. Figure 2 from the 1881 French *Album de Statistique Graphique* (Ministère des Travaux Publics, Plate 1) displays the tonnage transported by railroads across France in 1879, color-coded by region to distinguish railroad lines emanating from Paris. The lines of variable thickness show quantities of tonnage, large



Figure 2. Map showing the amount of railroad freight transported across France in 1879 along several regional routes (Ministère des Travaux Publics, 1881, Plate 1).

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(thick) and small (thin), and extend to every department and virtually every town across the entire country, personalizing the data for each French reader.

Designers also innovated within conventional plot frames—for example, snaking bars back and forth to make the data fit the genre, as shown in the bar chart in Figure 3 (lower arrow) from the 1883 *Scribner's Statistical Atlas of the United States* (Hewes & Gannett, Plate 123). Here, the production per capita of tiny Rhode Island exceeds the limits of the x-axis, so the bar impishly curls around and heads back in the other direction. The line graph in the lower part of the plate shows the cost of cotton, which, during the Civil War, literally went through the roof, breaking through the plot frame and extending all the way to Lake Michigan in the map above (upper arrow). This graphical disruption must have startled readers as well as posed a painful reminder of

the economic hardship the war inflicted—and relief that some things had returned to normal.

Pictorial elements: enargeia

Vivid visual description (or *enargeia*), both textual and visual, has long fostered emotional responses from audiences, and pictorial elements can serve that function in data displays, creating interest and excitement and personalizing the data. Although today pictorial elements are often viewed as superfluous “chartjunk” (Tufte, 1983), they had no such stigma when they first appeared in the late nineteenth century. In *Mulhall's Dictionary of Statistics* (1884) Michael Mulhall, one of the leading innovators, used picture charts to compare nation-states on their agriculture, shipping, steam power, and military weapons, a technique that appeared in *Peale's Home Library of Useful Knowledge* (Peale, 1886, p. 363), with several of the same kinds of pictorial charts occupying a single plate (Figure 4). Although using area for visual comparison lacks perceptual integrity,

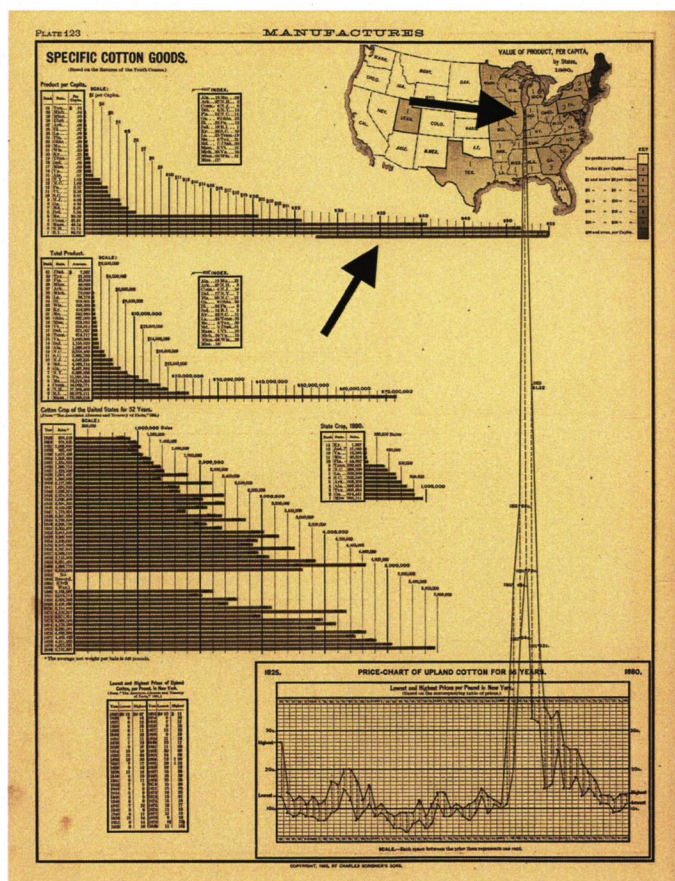


Figure 3. Novel breaking of boundaries (indicated by arrows) in the 1883 *Scribner's Statistical Atlas of the United States* (Hewes & Gannett, Plate 123). Courtesy of the Library of Congress, Geography and Map Division.

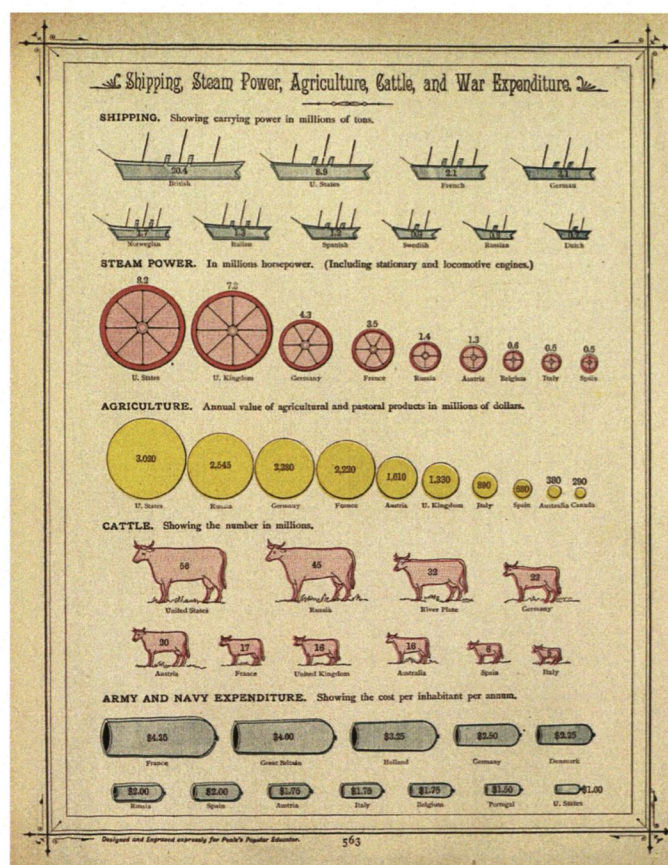


Figure 4. Pictorial charts from Richard Peale's 1886 *Home Library of Useful Knowledge* (p. 363).

Mulhall's pictorial method effectively drew readers into the data through visual stimulation and excitement, especially British and American readers, for whom the charts validated their global dominance (see Kimball, 2016). Intensive pictorial representations of data reached their zenith in the *Scientific American Reference Book* (Hopkins & Bond, 1913), which envisioned an array of U.S. national data in novel and sometimes fantastic designs using volume and area to show manufacturing, agricultural, government, and other measurements.

Epic narratives

Many displays in the golden age visualize vast narratives that tell compelling "stories," as Tufte puts it (1990, pp. 37, 108), about people and events, like the chart in Figure 1 that narrates the immense growth of a nation since its founding and the "Carte Figurative" of Minard (1878) visualizing Napoleon's Russian campaign. Minard earlier pioneered the graphical technique of using line thickness on maps to show variable trade routes, a display method echoed in the massive railroad map in Figure 2. In Minard's Napoleon chart, the shocking loss of life that he visualized with thick black bars receding from Moscow tell a foreboding story about foreign military adventures as well as serve as a grim memorial to his contemporaries who served on this calamitous campaign. Epic charts like Minard's and those of the U.S. statistical atlases visualized national narratives that evoked a wide range of emotional responses.

Collectively, data design during the golden age engaged readers emotionally by engendering curiosity, excitement, surprise, and national pride. These displays deployed color schemes and a plenitude of design elements typical of the Victorian age, which embodied the legacy of Romantic values aimed at arousing powerful feelings. Moreover, many of these displays told compelling, even epic, stories about historical events, population movement, economics, crime, and disease, often in a nationally competitive framework that revealed the audience's strengths and vulnerabilities.

Modernist Minimalism and the Emphasis on Functionality

The rich, inventive design of the late Victorian era that aroused the senses and the imagination was subsequently subdued by early modernism's emphasis on simple, functional design, stripped of all embellishments.

Early twentieth-century modernism coincided with a dearth of innovation in data design (Friendly, 2008, pp. 529–530), at least compared with the golden age that preceded it. Although this hiatus can be attributed to many causes, modernist aesthetics certainly played a role, with its emphasis on simplicity, stark functionalism, and perceptual immediacy. Modernism aimed to wring out emotion in all forms of design—architecture, painting, furniture, graphic design—by emphasizing objectivity and universal, geometrical forms that audiences could grasp without the mediating influences of culture, aesthetics, or sentiment.

Through this minimalism and emphasis on structural transparency, strict rationality (logos) regained its dominance over data design, clearly evident in the U.S. statistical atlases of the early twentieth century where a small range of data displays replaced the intricate, colorful forms of earlier atlases. Figure 5 shows two minimalist pie charts from the 1914 *Statistical Atlas of the United States* (Sloan, Plate 212.1, detail) that display data about the growing influx of immigrants, a hotly debated topic at the time, the passion about which is completely absent in these charts. Subsequently, minimalist charts like these appeared fairly routinely in U.S. Government reports (see, for example, U.S. Works Progress Administration, December 1937, p. 14; Committee on Interstate and Foreign Commerce, 1953, pp. 149–179). This modernist emphasis on simple, highly perceptible design was epitomized by the Isotype pictographs of Otto Neurath (1939) with their flat, machine-like forms, a style brought to the U.S. by Rudolf Modley (1937; see Crawley, 1994).

Later graphical theorists continued to advocate a highly functional approach that emphasized the connection between image, eye, and brain. For Jacques Bertin (1981), visual "information-processing" through perceptual efficiency provides the foundation for effective design, as graphical particles are manipulated to reveal patterns that can be *seen* (rather than read). Tufte (1983) also strives for perceptual efficiency with his "Lie Factor," "data-ink" and "data density" formulas, and his exhortations against "chartjunk" (pp. 57, 93, 107, 162), largely embracing what Lee Brasseur (2003) calls the "perceptual cognitive-based school of thought" (p. 4; see also Zachry & Thralls, 2004, p. 453). In this rational, functional approach to design, emotional elements serve as a decorative distraction that threatens to compete with or even corrupt the data.

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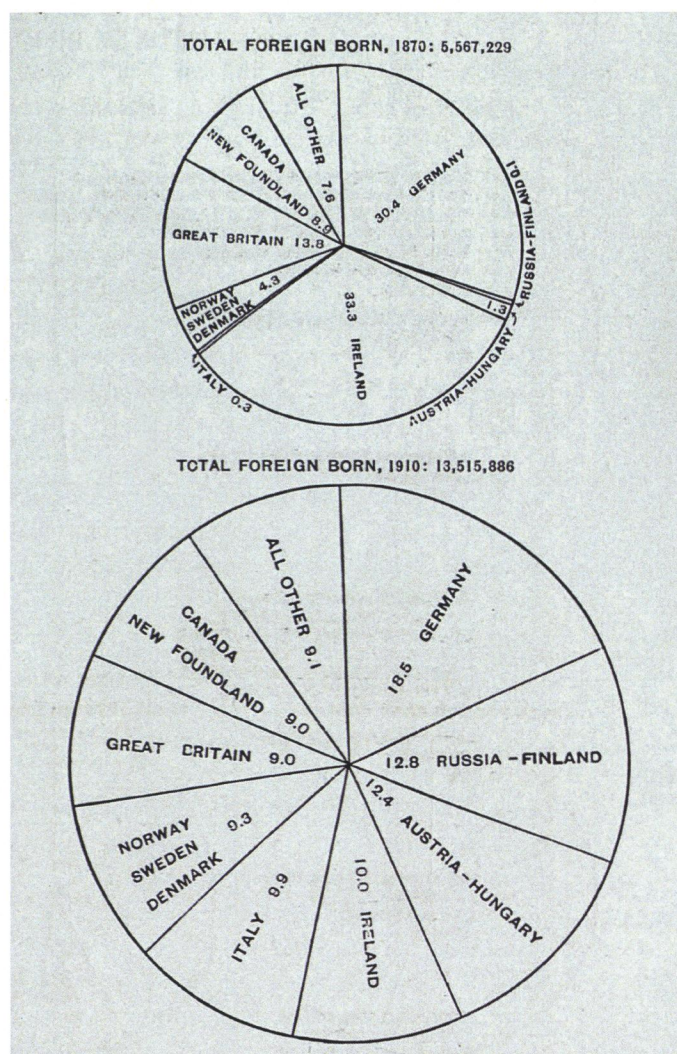


Figure 5. Plain pie charts from the 1914 *Statistical Atlas of the United States* showing the size and nationalities of the foreign-born population in two censuses (Sloan, Plate 212.1, detail).

Emotional Aspects of Contemporary Data Visualization

Despite the anti-pathos proclivity of modernism, emotional elements were not completely expunged from data design. Designers like Nigel Holmes (1984) in the popular media reinstated emotional elements—through 3D pictorial elements, vivid color, and hyperbole—cleverly embedding people, buildings, landscapes, and other objects into line, bar, and pie charts. However, these designs remained outside the realm of serious statistical graphics, dismissed by Tufte (1983) as bombastic hype aimed at perking up listless readers,

though his argument about feckless “chartjunk” has been challenged by Bateman et al.’s empirical study (2010).

Nonetheless, several leading theorists of contemporary data design have acknowledged some aspects of emotion or subjectivity in data design, if not argued explicitly for them. In the larger realm of document design, for example, Karen Schriver (1997) observes that “Readers’ interpretations of documents are shaped by thinking and feeling, by the subtle interplay of cognition and affect” (p. 189). More specific to charts and graphs, Edward Tufte, despite his perceptual/cognitive bent, occasionally makes nods to the “wonder,” “beauty,” and “graphical elegance” of data design (1983; pp. 121, 137, 177), and he promotes what Stephen Few (2012, pp. 295–306) and Nathan Yau (2013, pp. 61–68) describe as the “storytelling” capacity of data displays. Lee Brasseur (2003) critiques the cognitive/rational approaches, which she traces to the Cartesian spit between mind and body, asking pointedly, “Can good decisions be made about graphs which deny a place for emotion?” (p. 21). Sam Dragga and Dan Voss (2001), moreover, argue on ethical grounds for including emotive elements in data displays that show human suffering or loss as a way to “humanize” displays (p. 269). So several scholars, to varying degrees, have acknowledged or advocated emotional elements in data design at the onset of the digital revolution.

Although the modernist paradigm still dominates print displays, the digital age has rapidly departed from its stark functionalism. The technology of digital design and the era of “big data” have destabilized Tufte’s minimalist maxims, if not rendered them increasingly irrelevant. In this second golden age of data design, pathos elements have experienced a remarkable but yet largely unrecognized resurgence that digital design has fostered in at least three ways:

1. It revives the sensory stimulation prevalent in the late nineteenth century, with color, novelty, animations, and other multimodal features, eliciting a physiological response long associated with pathos from Aristotle to Campbell to the present.
2. It implements Campbell’s principles of personal proximity, of situating the circumstances in the audience’s own neighborhood, by enabling users to shape the display, by increasing their temporal and spatial proximity to the data, and by enabling interaction with the designer and other users.

3. It fosters expressive design by appealing to the audience's aesthetic sensibility or inviting users to interact with personal data, further increasing proximity between user and designer.

All three of these approaches to data design elicit emotional responses from audiences, with the latter two largely new with interactive data visualization.

Arousing the senses with the new digital vocabulary

Digital technology gives designers an array of visual elements that can engage audiences through sensory stimulation and in doing so engender a range of emotions—from excitement and delight to fear and anxiety. The infusion of new visual language has injected emotion back into data design on several fronts, including full color, multimodal features like sound and animation, and novel forms.

Coloring data In online digital design, color appears ubiquitously, often across the full spectrum of RGB values—unlike print where the financial cost of CMYK color still constrains its use. Online color engenders emotional responses on at least three levels: physiological, aesthetic, and cultural (see Richards & David, 2005). Physiologically, color fosters an almost immediate response that differs between warm colors and cool colors, with the former causing excitement or anxiety and the latter tending to soothe the viewer (see Mackiewicz, 2007, p. 147, for a summary of research). Heat maps offer a classic example of such emotional swings, with red and orange suggesting intensity or danger and blue and green stability and moderation. The U.S. National Gas Price Heat Map on the GasBuddy.com website (Coupal, Toews, & Gasbuddy/OpenStore LLC, 2016) illustrates this phenomenon, with bright red representing high prices and dark green low prices around the country. Aesthetically, color can also arouse emotional responses by pleasing or exciting us, depending on how well hues complement each other in the same field of vision or differ to create emphasis. However, sometimes the defaults on popular graphing software create gaudy combinations of hues, disrupting our aesthetic sensibilities. Culturally, color also plays on the emotions by triggering conventional interpretations that associate colors with various organizations, causes, events, and situations. For example, red can evoke joy and festivity (Chinese culture) or caution and anxiety (red ink in the U.S.). The emotional rollercoaster of

cultural interpretations can vary widely based on where in the world the audience is located, a daunting dilemma for online data designers with global audiences.

So deploying color to represent data opens a floodgate of possible emotional responses that designers have to be constantly vigilant about. Indeed, the lack of color in online visualization calls immediate attention to itself, as if the visualization had relapsed into the vacuous minimalism of the 1950s. Figure 6 shows a bar chart from the Bureau of Economic Analysis (U.S. Department of Commerce, 2016) with color distinguishing each category. This U.S. Government chart differs from the census chart in Figure 5, partly because it introduces a full palette of color within the plot frame, and because it allows users to customize the design (more on that later). Although the colors are muted (perhaps reflecting the chart's bureaucratic ethos), and depending on how many categories from the left panel are chosen for display (up to 10 are possible), the chart has the capacity to arouse the senses and stir emotion, especially the colored bars that fall below the x-axis and signal economic woes. The subtle pathos of this display's color, however understated, is nearly unavoidable in an online interactive display with such rich data.

Animating data Historically, one of the most effective emotional appeals involves telling a story to the reader, engaging the reader in a narrative. Many digital data designs do that by featuring animations that show change over time or space, enabling audiences to track variations in data through motion—across a map, scatterplot, or other plotting area. These compact, dynamic forms of storytelling create a sense of immediacy and add drama to the display. Digitally, animations thread together a series of “small multiples” (Tufte, 1983; pp. 170–175) in the form of individual data snapshots in the same plot frame (rather than spread over a print surface; indeed, some animations would fill many print pages of small multiples). Although researcher Danyel Fisher (2010) discovered perceptual drawbacks of displaying data in motion, he also found the animation he studied “to be more engaging and emotionally powerful” than static displays (p. 337).

This is certainly the case for the chart in Figure 7, which shows one of the world's most famous data animations, the Gapminder software (Rosling, Rosling Rönnlund, & Rosling, 2015; see also Rosling, 2006). Gapminder displays economic, health, education, and other sociological data from nations around the world

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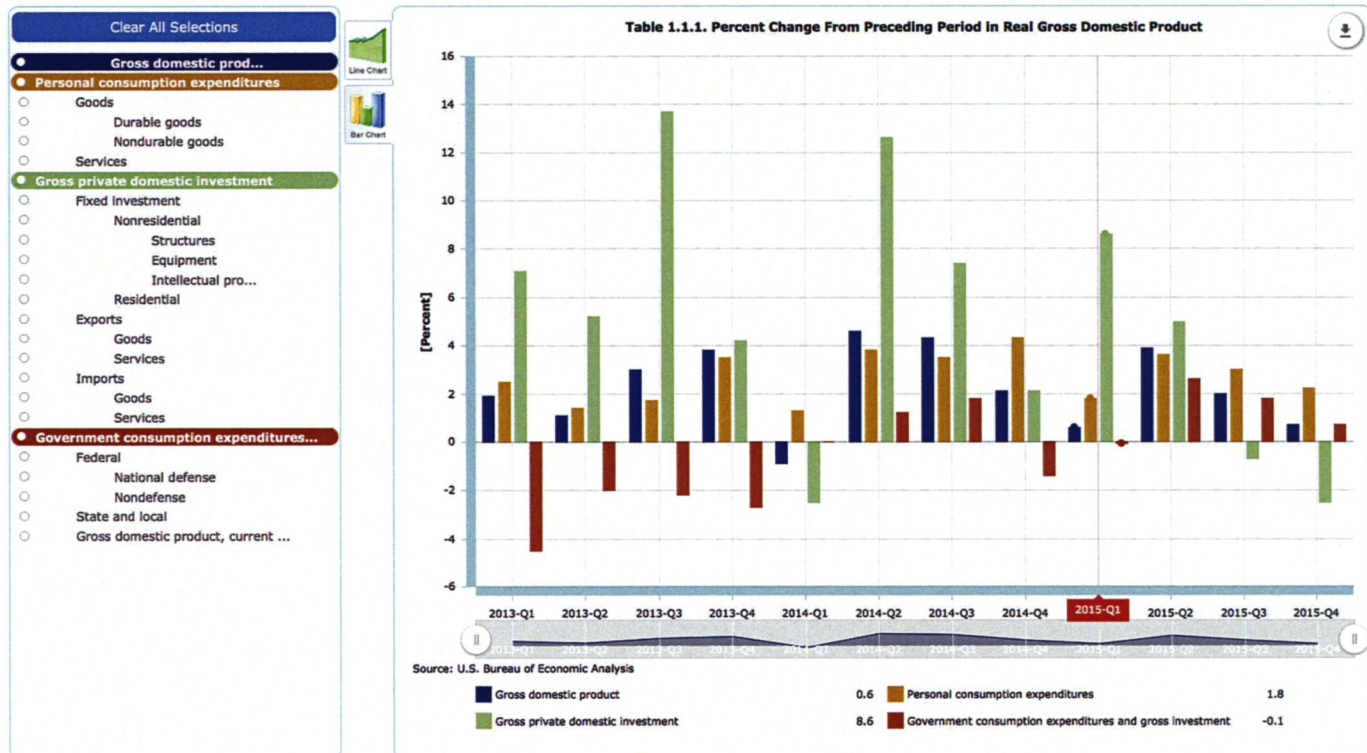


Figure 6. Interactive chart showing the percent change in parts of the gross domestic product by quarter of years (Bureau of Economic Analysis, 2016, Table 1.1.1). Courtesy U.S. Department of Commerce, Bureau of Economic Analysis.

on an animated scatterplot encompassing over 200 years, beginning in 1800. The Gapminder chart in Figure 7, for example, juxtaposes data about per capita income on the x-axis and longevity on the y-axis. As the story unfolds on this chart, nations color-coded by continent get larger, and as their economic conditions improve, their health typically improves as well, with the circles often bouncing around as they gradually drift to the upper right of the plot frame. As the narrative progresses, the animation fosters curiosity, excitement, and delight, revealing a great success story for most of humanity, though with some areas of the world lagging noticeably behind, heightening the drama by dampening optimism and fostering empathy.

The data animation in Figure 8, designed by Tim Klimowicz (2007), narrates grimmer data—deaths during the Iraq War. Echoing Minard's famous chart of Napoleon's Russian campaign, Klimowicz's design is a dramatic memorial to military personnel in the coalition who lost their lives in Iraq, displaying on the map vivid red bursts for each successive death as the conflict continued over several years, with each explosion

gradually fading into a black dot as successive bursts appear. The visual bursts of color are accompanied by the audible rattling of simulated gunfire that increases and decreases with the number of deaths represented visually. To focus the display, the viewer can select any of the nations in the coalition from the panel on the right. Although the designer wanted to remain "objective" (Klimowicz, Conclusion), this powerful multimodal display mourns and honors those losses, eliciting sadness and shock and depending on the viewer's connection to this conflict, grief, pride, pity, bitterness, or indignation.

Reinventing genres Invention and creativity play a large role in many online digital displays, resulting in novel forms that often combine or redefine existing genres. As we've seen, nineteenth-century readers were constantly scrutinizing new forms and observing the breaking of boundaries. The new online hybridization also creates curiosity and surprise that may be initially disorienting. Socially constructed genres are tenacious and stabilize user experience, especially in print, where they are most likely to be respected. Digitally, however, genres are often reshaped, experimented with,

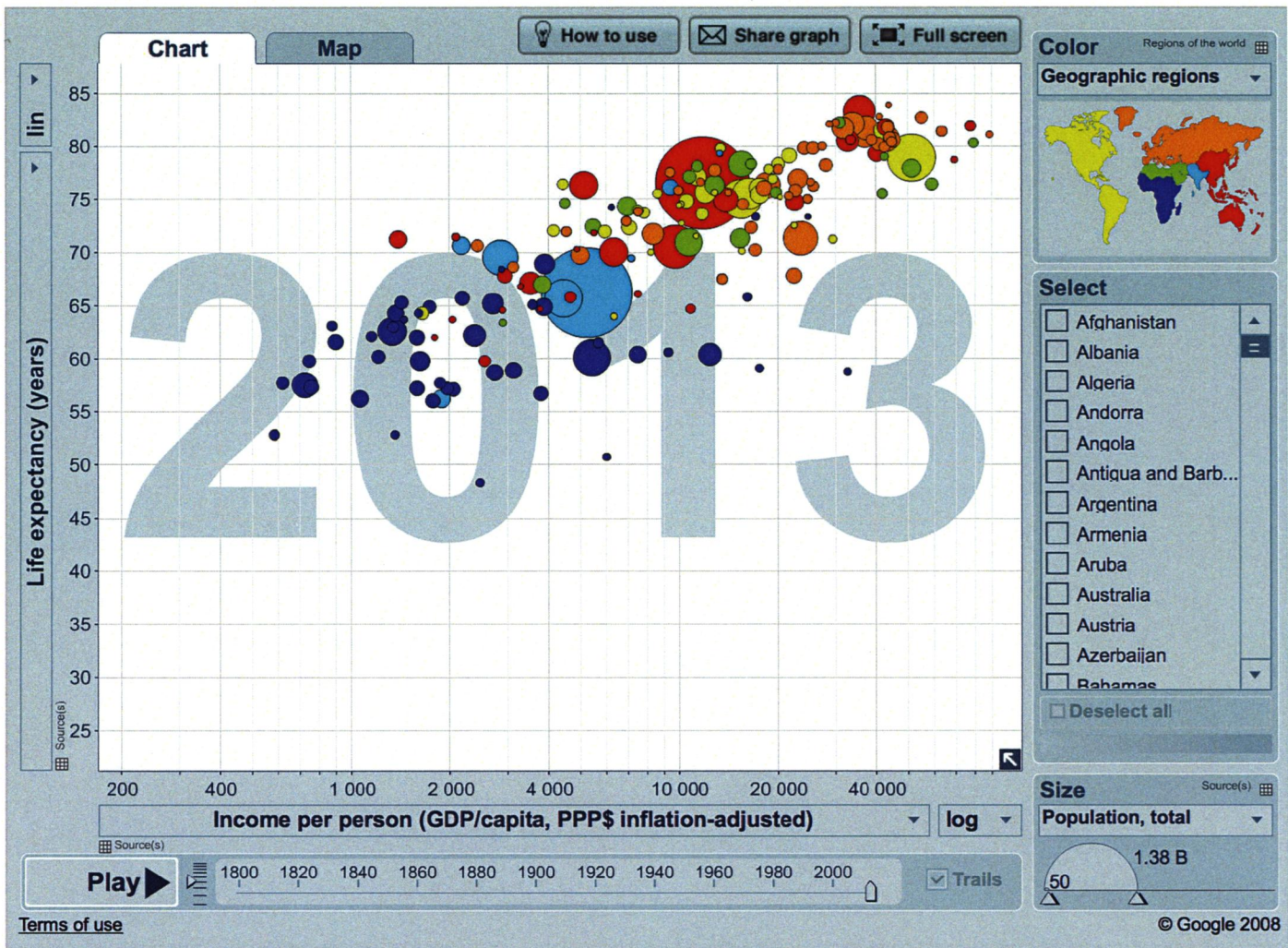


Figure 7. Gapminder animated scatterplot showing the relationship between life expectancy and income from 1800 to 2015 for people in countries around the world (Rosling, Rönnlund, & Rosling 2015). Source: Free material from www.gapminder.org

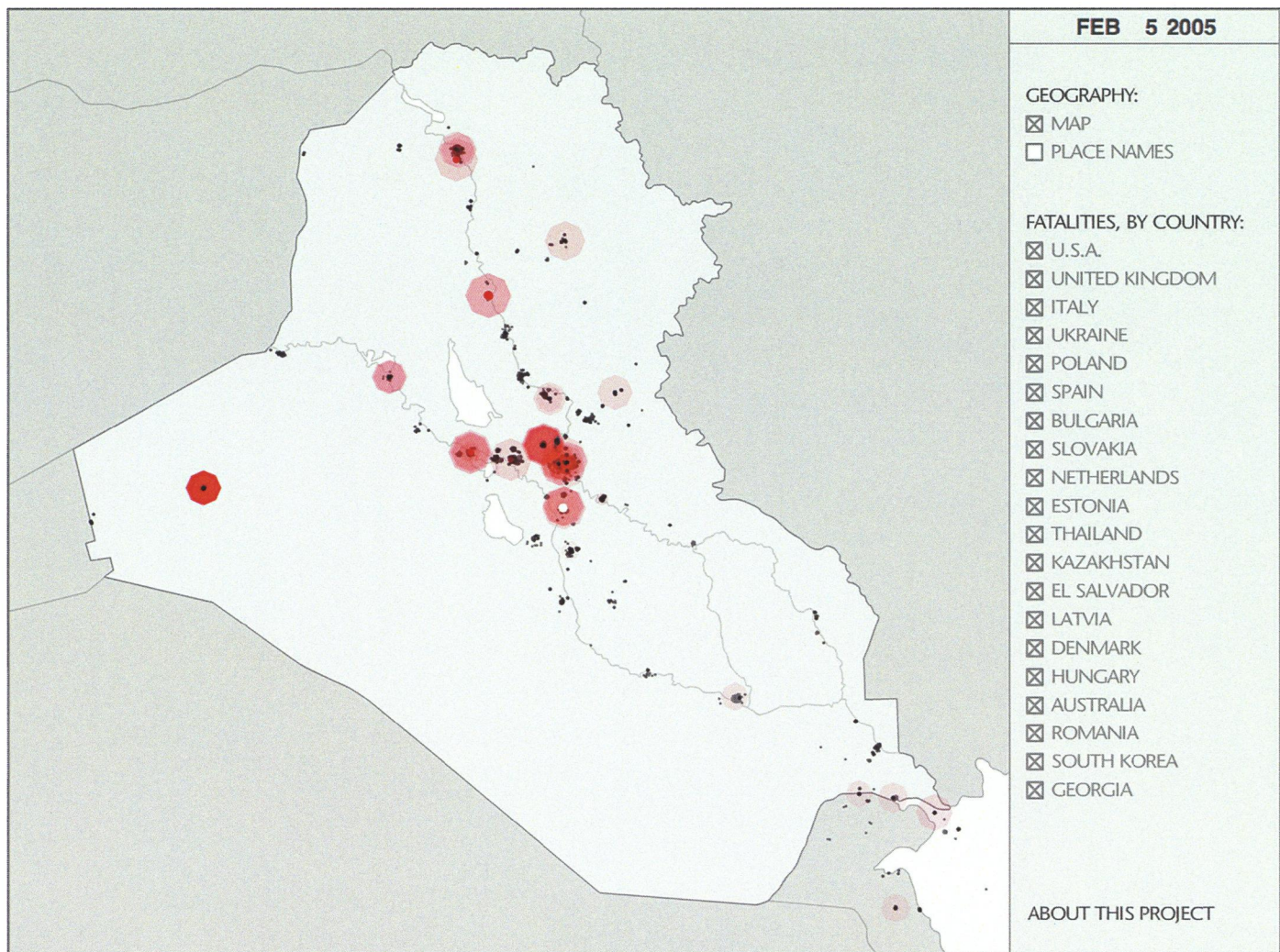
juxtaposed, and blended into hybrids. For example, treemaps are a contemporary form of mosaics, which were invented in the nineteenth century. New forms like word clouds, parallel sets (Kosara, 2016), and flower charts (Organisation for Economic Co-operation and Development, 2016) are not infographics designed to amuse but serious attempts to display complex data. The Gapminder chart in Figure 7 contains novel and hybrid elements, where area (the size of the circles) is integrated into a scatterplot (to show the relative population of each nation), color locates nations geographically, and animation narrates the data, with the years appearing successively within the plot frame. Many readers who first encounter this display have never experienced anything like it, evoking surprise, excitement, and

perhaps some initial bewilderment. The Iraq fatality chart in Figure 8, which is essentially a thematic map, elicits similar emotional responses through novel elements—the bursts of red, clicking sounds, and rapid movement through time.

Personalizing data through close proximity

Interactive data displays draw the audience closer to the data, fostering more intimate engagement. As George Campbell (1776) asserted in the eighteenth century, audiences care about, and are likely to make an emotional investment in, people and events nearby them, as opposed to those distant in space and time. Interactive data visualization achieves this proximity with audiences by enabling them to customize displays, by providing

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Up to date as of October 5, 2007.

Figure 8. Animated chart with sound designed by Tim Klimowicz (2011) that visualizes coalition fatalities in the Iraq War from 2003 to 2007. Reproduced with the permission of Tim Klimowicz.

visual access to real-time data nearby them, and by enabling them to participate in communities of users.

Custom displays: personalizing your design

Interactive data design changes the relationship between audience and visualization by enabling the audience to choose which data to visualize and often in what form. In this way, users can envision data at both what Tufte (1990) calls the “macro” level and “micro” level (pp. 37–51), or what Barton and Barton (1993) refer to as the “synoptic” and “analytic” views. Viewing data at the macro-level (or synoptic view) enables users to see the big picture, while viewing data at the micro-level (or analytic view) enables users to customize the view, to zoom in to details in the user’s own neighborhood (see

Kostelnick, 2008, pp. 124–125). In short, interactivity allows audiences to personalize the data, visualizing what matters to them and hiding what doesn’t.

Interactivity also often allows audiences to personalize the display itself: its genre (bar chart, scatter plot, line graph) and its graphical language (background color, coding of lines and bars). In these ways, every user has—or is at least afforded the possibility of—a unique, individualized experience with the display, a level of user agency that far exceeds that of static displays, even though that agency can vary considerably from one display to another, depending on the nature of the display and its interactive features (see Rawlins & Wilson, 2014).

With each audience member allowed to explore privately, the individual's interests and desires dominate this kind of visual experience, creating an emotional connection with the display. That highly private, individual experience whereby the user zooms in on local areas of interest invokes one of Campbell's "circumstances" (p. 209) associated with emotional appeals—the proximity to an audience member's own world, to things important to that individual. This kind of personal engagement can occur with graphical displays on virtually any topic—from finance, education, and politics to crime, weather, travel, and health. The Bureau of Economic Analysis chart (Figure 6) allows audience members to choose from dozens of data sets to envision, and within each set to choose categories they wish to display, and within this display to choose which genre (bar chart or line graph). Gapminder also allows the audience to choose categories from a detailed menu of nations (on the right of Figure 7) and thereby to isolate individual nations as the animation unfolds, as well as to choose several other data sets available on the website for similar display. The Iraq War fatality chart (Figure 8) also allows the audience to choose nations from the coalition, focusing the graphical display on victims of special concern.

The potential to personalize displays is increased exponentially in the charts in Figure 9 from the website StockCharts.com (2016) that show the fluctuations of the Dow Jones Industrials Average over 10 days in January 2016. This relatively streamlined display consists of candlesticks showing daily movement of prices and vertical bars below showing market volume. Figure 9, however, reveals only a fraction of the available charting options, which include the ability to display several technical indicators with areas, lines, and other graphical elements. Audiences can personalize the display by customizing the data (numerous time periods and data variables) and by customizing the graphical display. They can choose from over a dozen chart genres and from dozens of color schemes for the plot frame itself, depending on the mood the audience prefers, including the "Default" on the top of Figure 9 and "Night" on the bottom. Personalizing data design in these ways heightens the proximity between audience and display, intensifying the audience's stake in it and emotional connection.

Data dynamism and temporal proximity Most data displays, especially static paper displays, visualize phenomena that have already occurred—weeks, months,

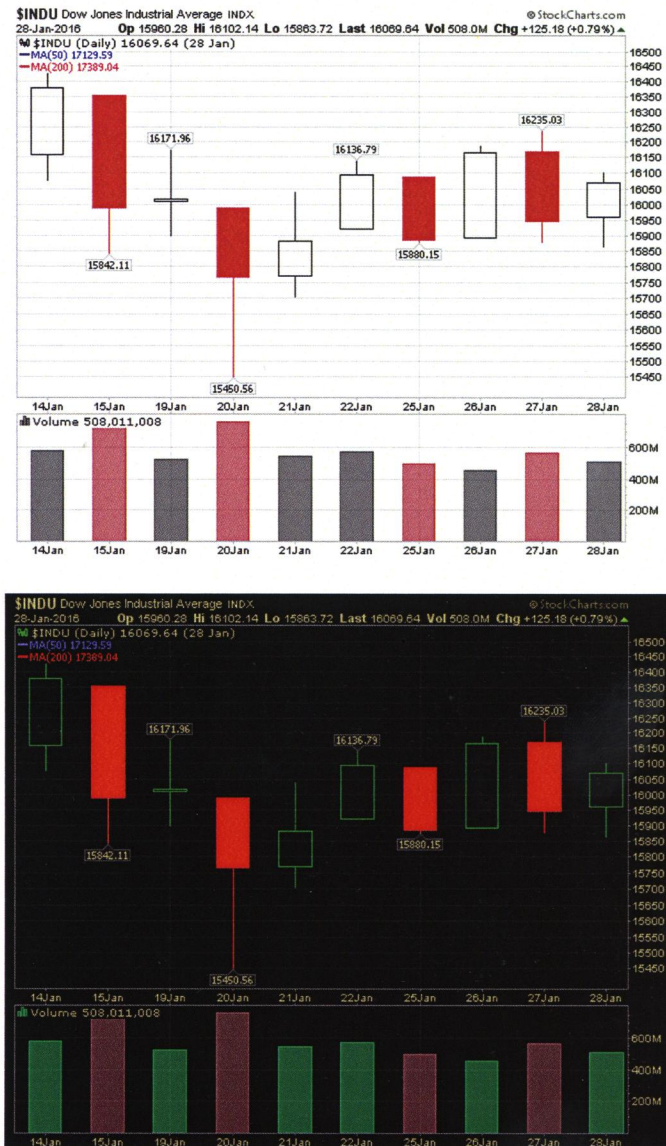


Figure 9. Charts from the StockCharts.com website showing the Dow Jones Industrial Average over 10 days in January 2016, with the default background (above) and with the background adjusted to "Night" mode (below). Copyright StockCharts.com. Charts courtesy of StockCharts.com.

or years ago. Many interactive displays, however, envision a constant flow of data, sometimes in real time, about markets (stocks, bonds, commodities), transportation (gas prices, highway traffic, airline flights), criminal activity, sports performances, and weather conditions and other natural phenomena. As data constantly refresh, moment-by-moment, and the graphical display changes accordingly, audiences feel

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connected to that display in the immediate present, as temporal proximity fosters an emotional bond.

When the personal stakes are high—especially when life, health, or property is implicated—the potential for fear and anxiety increases with real-time data. Displays that show fresh data about natural disasters or criminal activity, for example, can evoke feelings of anxiety, terror, or relief. However, even displays that envision more quotidian matters like investments, interest rates, gas prices, and traffic flow can engender delight, distress, or surprise. For example, the StockCharts.com chart in Figure 9 refreshes the Dow Jones Industrial average minute by minute, one of dozens of online charts that visualize stocks prices in near real time, creating the illusion of standing on a trading floor. For readers of these charts, the temporal immediacy and dynamism of visual data create an engaging experience fueled by constant anticipation. The dynamism of temporal proximity both energizes and enervates readers, depending on the corresponding emotions it engenders, ranging from anxiety or expectation, to joy or euphoria, to disappointment or dejection.

Too close for comfort: risk and fear Increasingly, online data visualizations are addressing large public audiences by alerting them to risks they face, ranging from natural disasters (earthquakes, hurricanes, sea-level rise) to crime to communicable diseases. These visualizations engender empathy if they are perceived as distant threats but arouse fear and anxiety when they are close at hand. As Aristotle pointed out, fear is aroused by that which “has the potential for great pains or destruction, and these [only] that do not appear far off but near, so that they are about to happen” (p. 128). In other words, as Campbell later argued, both spatial and temporal proximity have the capacity to arouse the emotions. Because contemporary visualizations typically allow users to customize the display to focus on their immediate location, that close proximity triggers and heightens the emotional response. To residents of Tampa, an animated warning map tracking a typhoon in the Philippines can engender empathy, but a similar animation of a hurricane wandering across the Gulf of Mexico threatening their lives and property will induce fear and anxiety. Other visualizations, at varying degrees of abstraction, can engender similar responses:

- heat maps that deploy shades of red, orange, and yellow to show impending dangers like severe storms, tornadoes, or forest fires;

- crime maps that identify at street level various kinds of crime with icons or that provide pictures of local sex offenders; and
- risk maps that show the probability of an earthquake, volcanic eruption, or flooding in a given location.

In all of these examples, because of the proximity of the danger that threatens them or their loved ones, audiences become engaged emotionally in the visualization.

For example, interactive maps created by the National Oceanic and Atmospheric Administration (NOAA, 2016) visualize the risk of sea-level rise along coastal areas in the U.S. Figure 10 shows a risk map for flooding along the North Carolina coast. By clicking on the icons, users can see a picture of a landmark building in the area and can select a water depth to see the effects on that structure, which in Figure 10 is a lighthouse. This dramatic picture provides descriptive detail, or *enargeia*, that warns readers of the devastating effects of rising waters at varying depths from one to six feet. Pictures like this one appear on NOAA’s sea-level rise map across the U.S. Atlantic, Gulf, and Pacific coasts, raising audience awareness by visualizing familiar places—docks, streets, bridges, parks, and public buildings—inundated with floodwaters. Proximity fosters pathos: the emotional trauma an audience experiences by seeing its own neighborhood (or property) underwater, however imminent or hypothetical, is hard to measure. On the other hand, knowing that a house or business escaped a natural disaster can surely pacify anxious readers.

Risk from human threats is visualized in Figure 11, a crime map of Chicago, Illinois, that appeared on the CLEARMAP: Crime Incidents website, created by the Chicago Police Department (2016) to visualize criminal activity across the city. The interactive map in Figure 11 displays crimes in a Chicago neighborhood over two weeks, using descriptive icons for each type of crime: a gun for aggravated assault and battery, brass knuckles for simple assault and battery, a mask for robbery, a figure tampering with a car for auto theft, and a sprawled body for homicide and manslaughter. The interactive map on another crime mapping website, Crimemapping.com™ (The Omega Group, 2016), uses a mask for robbery, a clenched fist for assault, and a fractured hood/windshield for a car broken into. Yet another crime mapping website, SpotCrime.com (2016), uses

crosshairs for a shooting, a figure in a dark clothes for burglary, and a figure toting money bags for robbery. The icons on these interactive crime maps vary in their level of abstraction, analogy, and hyperbole, and, in doing so, might heighten (or belie) the gravity of the data they represent, eliciting a variety of emotions. For victims of these crimes, or those who know the victims, the map could be a painful reminder of the event or provide comforting reassurance that justice will prevail; for those who live in a given area, the map is an anxiety-inducing warning of dangers that lurk—or emotionally empowering if it fosters vigilance and control.

Conversations over the digital fence: user/designer interaction In the larger realm of digital

technology, James Zappen (2005) describes a “new digital rhetoric that encourages self-expression, participation, and creative collaboration” (p. 321)—activities in interactive data visualization that arouse emotion and differ sharply from the ways audiences previously interacted with data displays. Traditionally, most charts and graphs have appeared to their audiences as fixed entities—final, reified artifacts that manifest what Walter Ong called the “closure” of print (1982, pp. 132–135)—whereby audiences individually analyze and evaluate them in isolation from the designers or other audience members. As a result, audiences for static displays largely forego any particular attachment to the display or its creator; in contrast, interactive displays

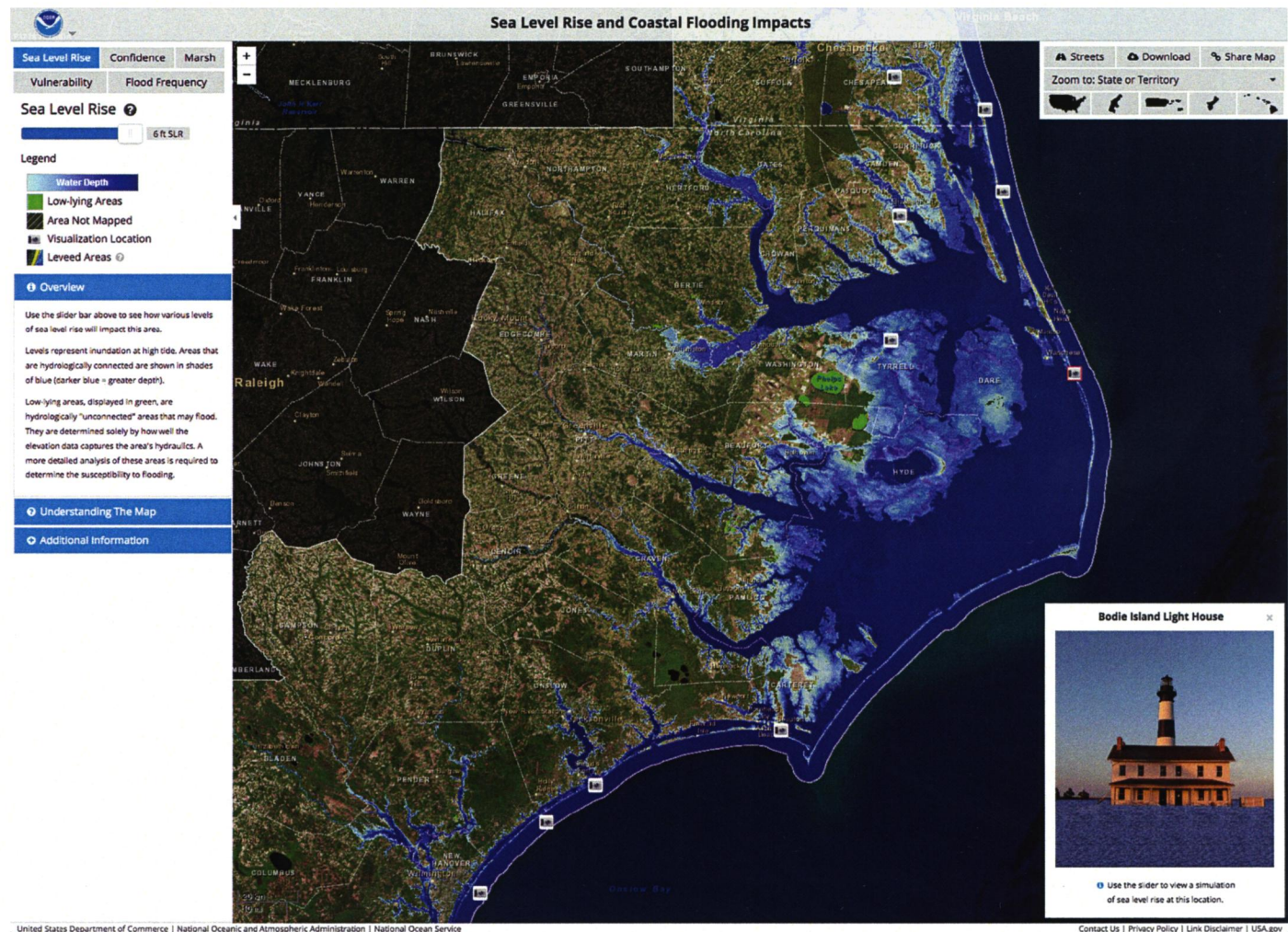


Figure 10. Interactive map showing the likely effects of 6 feet of sea level rise on the coast of North Carolina, with a picture illustrating the consequences for a local lighthouse (National Oceanic and Atmospheric Administration, 2016). Figure courtesy of NOAA. Used by permission. Copyright © 2016 Esri and its data providers. All rights reserved.

- a Contact link, whereby audience members can provide confidential commentary directly to the designer, and the designer can respond to it.

Through proximity, these affordances personalize the interaction between designer, display, and audience, replacing the distant and sterile relationship of static print displays with a dynamic, familiar one that invests the audience emotionally. All of the interactive displays discussed in this article afford, at a minimum, some level of contact between audience and designer (or the organization that sponsors the design) or with other related agencies. For example, the StockCharts.com chart (Figure 9) has a “Report Problems” link beneath the chart that enables users to critique or comment on a given chart, and on a SpotCrime map, mousing over a given crime icon provides details about the case and enables users to access the local police department, where they can ask questions or provide information.

Expressive displays: visualizing the self

Emotional appeals also include expressive elements that arouse an audience’s aesthetic sensibility or, like expressive writing, originate from within the individual, giving a communication voice and authenticity. Expressivism in data visualization encompasses both of these aspects through designs that show qualities of beauty, or otherwise appeal to the audience aesthetically, and designs that give audiences glimpses into the designer’s personal life, increasing the proximity between audience and designer.

A growing mode of expressive data design is what Nathan Yau (2013) calls “data art,” a form of self-expression “where the imagination runs wild, data and emotion drive together, and creators make for human connection” (p. 74). The goal of these expressive designs is “to experience data, which can feel cold and foreign” (Yau, p. 74)—in other words, for data design to realize its capacity to arouse the emotions. Lev Manovich (2013) acknowledges the capacity of contemporary “information-visualization designers” to “evoke particular emotions in the viewer” (pp. 12–13), and Noah Iliinsky (2010) touts the novelty and beauty of contemporary data design and the excitement it offers its audiences. Expressive “data art” has almost unlimited forms, which Yau illustrates in two- and three-dimensional designs visualizing phenomena as varied as the overhead skies, athletic performances, and dating websites (pp. 74–84). Although some of these designs might be more poetic

than functional, and more appropriate for an exhibition gallery than a PowerPoint, they illustrate the expressive capacity of digital design, which even the serviceable financial chart in Figure 6 can’t completely contain.

Expressive data design is also occurring in the growing trend to “quantify” the self, whereby designers collect and visualize data about their daily lives. To encourage personal data collection, Quantified Self Labs organizes meetings and forums and offers “a guide to self-tracking tools” for individuals interested in monitoring data about themselves (Wolf, Kelly, & Quantified Self Labs, 2015, “About the Quantified Self”). In this rapidly growing movement, designers visualize data about their health, work activities, finances, travel, and social interactions and often post their designs online. Exemplars of these kinds of displays have appeared on the Internet as personal annual reports, such as those of designer Nicholas Felton (2015) and of programmer/data designer Jehiah Czebotar (2015).

Figure 12 shows an interactive donut chart that appears in Czebotar’s 2014 annual report. The display visualizes his visits to coffee shops in the New York City area over a year, with each segment of the donut having an interactive feature that displays a bar chart showing

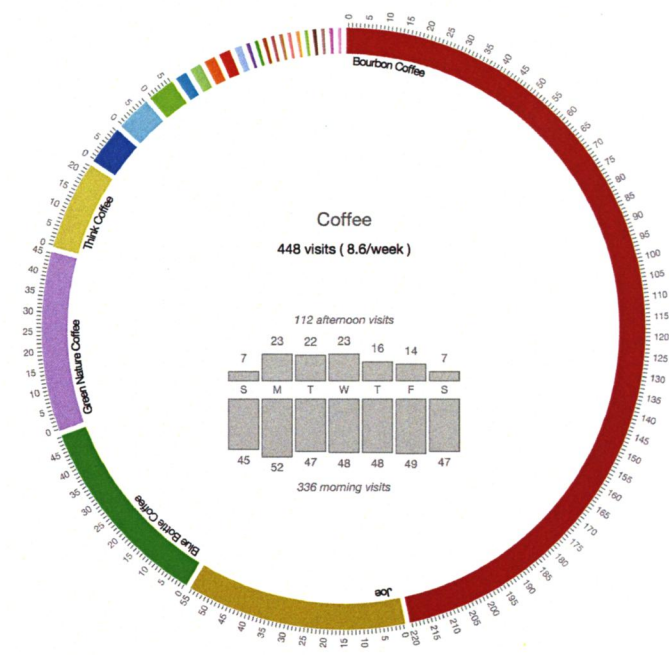


Figure 12. Interactive chart from Jehiah Czebotar’s 2014 personal annual report that shows time spent at several New York City area coffee shops. Copyright 2016 Jehiah Czebotar. Reproduced with permission of Jehiah Czebotar.

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visits to that particular coffee shop each day of the week, divided into mornings (bottom) and afternoons (top). Visualizing self-collected data reveals the private activities of designers to anyone who wishes to observe them, a form of data design that should not be dismissed as mere self-indulgence. Like expressive writing, these designs originate entirely from within the self but also have functional and emotive value by giving users insights into behavioral patterns that they might share and prompting them to consider (and perhaps quantify) other such patterns in their own daily lives.

Benefits, Drawbacks, and Implications of Envisioning Emotion

Given this new data design landscape, what are its benefits and drawbacks and what are the implications for practice? These new forms obviously have great power to engage, persuade, and motivate audiences by eliciting emotions ranging from surprise and delight to empathy and fear. By entering the domain of the subjective, irrational, and personal, these displays can wield potent and perhaps unprecedented rhetorical power.

As such, designers and their audiences stand to accrue many benefits from data visualization. Readers gain the ability and confidence to explore complex data sets that might otherwise remain hidden or inscrutable to them, and they can do so under their own control, a level of micro-level accessibility that personalizes the experience. Access to customizable data, sometimes real-time data, can also give audiences a sense of empowerment for decision-making, even in the face of anxiety-inducing risks. Audiences also have the opportunity to critique a given design, query its creator, or interact with other audience members, socializing the visualization among a community of designers and users.

On the other hand, online data visualizations can place interpretive demands on readers who might be moved emotionally by design features but puzzled by their lack of obvious functionality. The richness of digital graphical language, with full color and animation, along with other multimodal features, can create a fine line between engagement and distraction. Moreover, the increasing trend of users to access displays on small screens (tablets, smartphones) poses challenges for data-rich, multimodal designs with interactive features that might not be as perceptually or emotionally compelling as they appear on laptop or desktop screens.

The emotional aspects of contemporary data visualization have several implications for designers:

1. Deploying color and multimodal features elicits a wide range of emotional responses from their audiences at both a physiological and a cultural level that can be used to focus attention and persuade.
2. Enabling users to customize displays at a micro-level can make the displays more personal and satisfying. Also, micro-level details in the form of data labels and descriptions can compensate for perceptually challenging designs that rely on area, circles, or gradients. In addition, providing a downloadable Excel data file will further increase user interactivity with the data as well as user trust.
3. Giving the audience opportunities for feedback and discussion socializes the display, enhancing audience engagement and understanding. Contact information, a comment/message board, or a link to a blog or social media can provide these opportunities.
4. Displaying fluid and even real-time data increases temporal proximity, heightening the emotional appeal and credibility of the display. Connecting the display to an active database, including a date on the display, and building an archive of previous visualizations can bolster user confidence in its timeliness.
5. Integrating expressive elements, however modest, can personalize the designer's voice, drawing readers into the data and creating a stronger relationship with the display. Thematic color or pictorial elements, data-bearing icons, and designer bios or commentaries can provide some options without devolving into "chartjunk."

Also, interactive data visualization will likely reshape some of the design conventions of traditional paper displays, changing audience expectations. Simulating in print some interactive conventions (color, micro-level features, designer/user interaction) might also heighten the emotional appeals of static print displays.

Conclusions

Emotional appeals are rapidly being reintegrated into data design, changing reader expectations about visualizing increasingly larger and more complex data sets. Like the explosion in innovation in the late nineteenth century, the new golden age of data visualization offers a feast for the senses, stirring the

emotions through ubiquitous color, novelty, and multimodal features. As the shift from modernist minimalism to dynamic digital displays continues, the surge of emotional appeals will only intensify, impelled by the personalizing effects of interactivity, spatial and temporal proximity, and expressive design.

Understandably, given the proclivity to view data design as an entirely rational process, some designers (and users) might be reluctant to accept—indeed strongly resist—the notion that data design might be enhanced by emotional appeals, let alone that emotion may already infuse displays that they construct or interact with. Like any form of information design, however, the new visual language of interactive data design must continually be scrutinized. Designers who understand this new language—and its rhetorical power—can successfully deploy and control it in order to make complex information more humane, relevant, and usable.

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